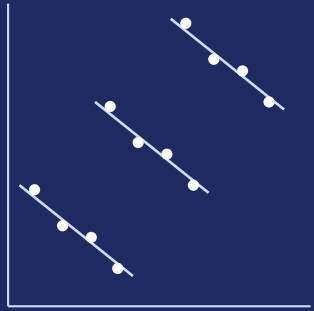


RANDOM VS. FIXED EFFECTS

Random vs. Fixed Effects

Class 2 · Hour 1

Prof. Dr. Rodrigo Leite
Panel Data · EBAPE/FGV

CLASS 2 · HOUR 1

What We'll Cover This Hour

01

The fixed effects model

Each unit gets its own intercept: the within transformation and LSDV

02

The random effects model

Treating the unit effect as a random draw, and the GLS logic behind it

03

The assumption that separates them

Whether the unit effect is correlated with your regressors

04

Estimating all three in Stata

Pooled OLS, RE, and FE on the same data

Recap: Where Pooled OLS Left Us

Last class: pooled OLS is consistent only if the unit effect α_i is uncorrelated with x_{it} — an assumption that often fails in practice.

> Today we stop *hoping* the assumption holds and **model the unit effect directly**, in two different ways.

Fixed effects

Treats α_i as a **parameter to eliminate**. Works whether or not α_i is correlated with x.

Random effects

Treats α_i as part of a **random, uncorrelated error**. More efficient — but only if that assumption holds.

The Fixed Effects Model

$$y_{it} = \beta_1 x_{it} + \alpha_i + \varepsilon_{it}$$

α_i is now a **parameter** — a separate intercept for each firm, person, or country — not part of the error term. This is what lets FE absorb any time-invariant confounder, observed or not.

Also called the *within* estimator, or LSDV (least-squares dummy variable) model.

Key assumption: strict exogeneity

$E(\varepsilon_{it} \mid x_{i1}, \ldots, x_{iT}, \alpha_i) = 0$: shocks in any period are unrelated to x in *every* period.

What it rules out

Feedback from past shocks to future x, and lagged y as a regressor (dynamic panels come later in the course).

How FE Is Estimated: The Within Transformation

Subtract each unit's own average from every variable — the unit effect cancels out algebraically.

$$\underbrace{(y_{it} - \bar{y}_i)}_{\ddot{y}_{it}} = \beta_1 \underbrace{(x_{it} - \bar{x}_i)}_{\ddot{x}_{it}} + (\varepsilon_{it} - \bar{\varepsilon}_i), \qquad \hat{\beta}_1^{FE} = \frac{\sum_i \sum_t \ddot{x}_{it}\, \ddot{y}_{it}}{\sum_i \sum_t \ddot{x}_{it}^2}$$

1	α_i **disappears**	It is constant within unit i, so subtracting the unit mean removes it: $\alpha_i - \bar{\alpha}_i = 0$
2	**Only within variation remains**	The estimator uses deviations from each unit's own average over time
3	**Time-invariant regressors vanish too**	Any variable that never changes for a unit is wiped out along with α_i
4	**Degrees of freedom**	Residual df is $NT - N - K$: the N means are estimated, even if implicitly

What Fixed Effects Actually Controls For

Any characteristic of a unit that is stable over the sample period — measured or not — is automatically absorbed.

Management quality

A firm's stable managerial talent or culture

Geography

A country's location, climate, or natural resources

Founding conditions

A firm's initial capital structure or founder identity

Time-invariant institutions

A region's legal origin or historical political system

The catch: "stable" means stable *over your sample*. Management quality that changes when the CEO changes is **not** absorbed.

Reading the Fixed Effects Coefficient

“When this unit’s x changes, how does this same unit’s y change, on average?”

A purely within-unit story — that is both its strength and its limit:

Strength

Immune to any confounder that never changes for that unit

Limit

Says nothing about why one unit differs from another on average

An Equivalent Route: LSDV

Add one dummy variable per unit and run OLS: the least-squares dummy variable approach gets the *identical* $\hat{\beta}_1$.

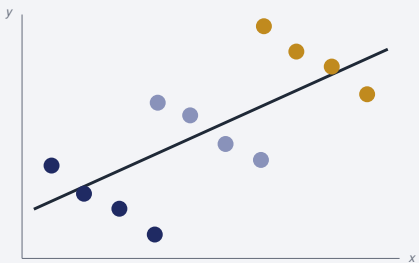
$$y_{it} = \beta_1 x_{it} + \gamma_1 D_{1i} + \gamma_2 D_{2i} + \cdots + \gamma_N D_{Ni} + \varepsilon_{it}$$

1	**Same $\hat{\beta}_1$, same residuals**	Mathematically identical to the within estimator (Frisch–Waugh–Lovell), just computed differently
2	**Costly with many units**	One dummy per unit means N extra parameters: impractical when N is in the thousands
3	**Why we usually demean instead**	Same $\hat{\beta}$ without estimating thousands of dummy coefficients
4	**Don't read the $\hat{\gamma}_i$ too closely**	With short T, each unit's intercept is estimated from very few observations

Back to Last Class's Puzzle: What FE Sees

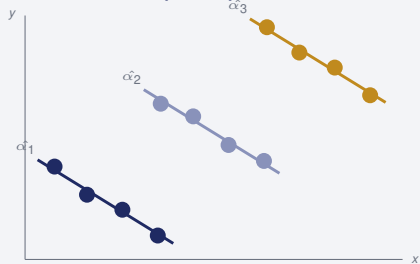
Same 12 points as the Simpson's paradox example. FE fits one line *per unit*, with a common slope.

Pooled OLS



One intercept, one slope: $\hat{\beta} > 0$

Fixed effects (LSDV)



Three intercepts, one common slope: $\hat{\beta} \approx -0.77$

Letting each unit have its own α_i removes the between-unit confound and recovers the true (negative) within relationship.

The Cost of Fixed Effects

1	**Time-invariant regressors drop out**	You cannot estimate the effect of gender, country of incorporation, or founding year alongside unit FE
2	**Only within-unit variation is used**	If x barely changes over time for each unit, FE has little variation left: large standard errors
3	**Efficiency loss**	Throwing away between variation costs precision relative to pooled OLS or RE
4	**Measurement error is amplified**	Demeaning removes signal but keeps noise, so attenuation bias can get worse
5	**Incidental parameters problem**	With fixed T, the $\hat{\alpha}_i$ are inconsistent. In linear models $\hat{\beta}$ is still fine; in nonlinear models (logit, probit) the problem contaminates $\hat{\beta}$ too

The Random Effects Model

$$y_{it} = \beta_0 + \beta_1 x_{it} + \underbrace{\alpha_i + \varepsilon_{it}}_{u_{it}}, \qquad \alpha_i \sim (0, \sigma_\alpha^2), \;\; \varepsilon_{it} \sim (0, \sigma_\varepsilon^2)$$

α_i is treated as a **random variable** and folded back into the composite error, much like pooled OLS. The difference: RE explicitly models the error's structure and is estimated by **(feasible) GLS**, not plain OLS.

RE assumption 1: no correlation with α_i

$E(\alpha_i \mid x_{i1}, \ldots, x_{iT}) = 0$ — the unit effect is unrelated to x in *every* period

RE assumption 2: strict exogeneity

$E(\varepsilon_{it} \mid x_{i1}, \ldots, x_{iT}, \alpha_i) = 0$ — the same condition FE needs

RE needs everything FE needs, *plus* assumption 1. That extra assumption buys efficiency.

The Intuition Behind RE's GLS Estimator

Random effects doesn't fully demean like FE: it **partially** (quasi-) demeans.

$$(y_{it} - \theta\bar{y}_i) = \beta_0(1-\theta) + \beta_1(x_{it} - \theta\bar{x}_i) + \text{error}, \qquad \theta = 1 - \sqrt{\frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2}}$$

1. $\theta \to 1$**: RE behaves like FE** — When the unit effect dominates ($T\sigma_\alpha^2$ large relative to σ_ε^2): many periods, or large unit-level variance
2. $\theta \to 0$**: RE behaves like pooled OLS** — When the unit effect is small ($\sigma_\alpha^2 \approx 0$): there is little to remove
3. θ **is estimated from the data** — Stata reports it with `xtreg ..., re theta`; in unbalanced panels it varies with each T_i

Three Estimators, Side by Side

	Pooled OLS	Random effects	Fixed effects
Treats α_i as	ignored (in error)	random, uncorrelated	parameter to remove
Estimated by	OLS	FGLS (quasi-demeaning)	OLS on demeaned data
Uses time-invariant x?	Yes	Yes	No
Uses between variation?	Yes	Yes	No
Consistent if $\text{Cov}(\alpha_i, x_{it}) = 0$?	Yes	Yes	Yes
Consistent if $\text{Cov}(\alpha_i, x_{it}) \neq 0$?	**No**	**No**	**Yes**
Efficient if RE assumptions hold?	No	**Yes**	No

The Assumption That Decides Everything

Everything comes down to one question: **is** $\text{Cov}(\alpha_i, x_{it}) = 0$**?**

Yes, uncorrelated

Both RE and FE are consistent. RE is **more efficient**, because it uses within *and* between variation.

No, correlated

RE is **inconsistent**. FE is the only one of the two still standing.

This is the logic of the **Hausman test** (next hour): if both are consistent they should agree; a large gap between $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$ is evidence against RE.

Efficiency vs. Consistency

Fixed effects is the safer default: consistent whether or not the RE assumption holds, at a cost in precision.

If the RE assumption holds

RE is consistent and more efficient: smaller standard errors, same target.

If the RE assumption fails

RE is inconsistent; FE remains consistent, just less precise.

Practical stance

Most applied researchers in business and economics default to FE unless there is a strong reason to believe the RE assumption.

RE remains attractive when the key regressor is time-invariant, or when x varies mostly between units.

Estimating All Three in Stata

```
clear all
webuse nlswork
xtset idcode year
regress ln_wage hours union age, vce(cluster idcode)      // pooled OLS
xtreg   ln_wage hours union age, re vce(cluster idcode)   // random effects
xtreg   ln_wage hours union age, fe vce(cluster idcode)   // fixed effects
areg    ln_wage hours union age, absorb(idcode) vce(cluster idcode) // LSDV
```

`xtreg ..., fe / re`	the within (FE) estimator / the FGLS random effects estimator
`areg, absorb()`	LSDV without printing 4,700 dummies; same $\hat{\beta}$ as `fe`, SEs can differ slightly (df adjustment)

In class: compare the coefficient on `union` across the models. What does the gap between pooled OLS and FE suggest about $\text{Cov}(\alpha_i, \texttt{union})$?

Check Your Understanding

1. **Dropped regressor** — You add a firm's founding year to an FE regression of ROA and Stata omits it. Why?
2. **Reading θ** — Stata reports $\hat{\theta} = 0.92$ after `xtreg, re`. Which estimator will RE resemble, and why?
3. **Disagreement** — $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$ differ substantially. What does this suggest about $\text{Cov}(\alpha_i, x_{it})$?
4. **Precision** — Why are FE standard errors often much larger than pooled OLS ones on the same data?

Recap and What's Next

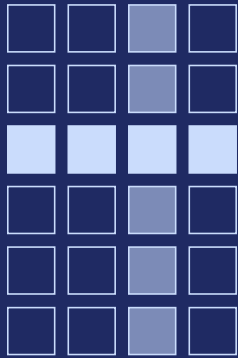
Takeaway: FE and RE both model unit heterogeneity directly instead of hoping it away. FE is the robust default; RE trades robustness for efficiency.

Next: Hour 2 — Breusch–Pagan, Hausman & Time Fixed Effects

Formal tests to choose among pooled OLS, RE, and FE — then adding time fixed effects for shocks common to all units.

Then: Hour 3 — paper discussions.

Questions?



BREUSCH–PAGAN · HAUSMAN · TIME FE

Testing & Time Fixed Effects

Class 2 · Hour 2

Prof. Dr. Rodrigo Leite
Panel Data · EBAPE/FGV

CLASS 2 · HOUR 2

What We'll Cover This Hour

01

The Breusch–Pagan LM test

Is there a unit effect at all, or was pooled OLS fine?

02

The Hausman test, in depth

The statistic, its assumptions, when it misbehaves, and robust alternatives

03

A complete testing sequence

Combining the tests into one estimator-choice workflow

04

Time fixed effects

Controlling for shocks common to every unit in a given period

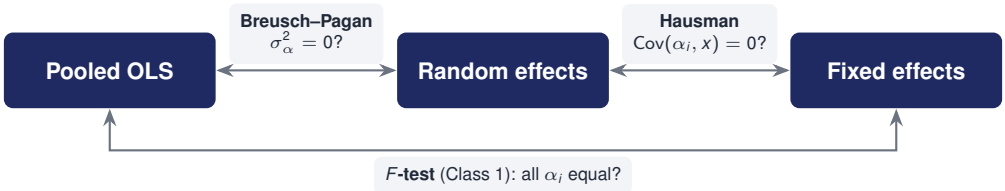
05

Two-way fixed effects in Stata

Syntax, interpretation, and testing whether time effects are needed

Recap: Three Estimators, One Open Question

Pooled OLS, RE, and FE handle α_i differently — but we have not yet asked which one the data support. Today we close that gap, and add a second source of confounding: **time itself**.



Where the Breusch–Pagan Test Fits

Before choosing between RE and FE, ask a simpler question: **is there any unit effect at all?**

> The Breusch–Pagan Lagrange multiplier (LM) test checks whether the variance of the unit effect is zero — that is, whether pooled OLS was fine all along.

No unit effect detected

Pooled OLS is not just convenient — it is supported by the data.

Unit effect detected

Move on and choose between RE and FE with the Hausman test.

The Breusch–Pagan Test: Hypotheses and Statistic

H_0: $\sigma_\alpha^2 = 0$ (pooled OLS is appropriate) vs. H_1: $\sigma_\alpha^2 > 0$ (a unit effect exists)

With pooled OLS residuals e_{it} (balanced panel):

$$LM = \frac{NT}{2(T-1)} \left[\frac{\sum_i \left(\sum_t e_{it} \right)^2}{\sum_i \sum_t e_{it}^2} - 1 \right]^2 \sim \chi_1^2 \text{ under } H_0$$

1. **Start from pooled OLS residuals** — Estimate the simple pooled model and keep its residuals; one restriction, so one df
2. **Check whether residuals cluster by unit** — If a unit's residuals share a sign, $\left(\sum_t e_{it} \right)^2$ is large relative to $\sum_t e_{it}^2$

You won't compute this by hand: Stata's `xttest0` does it after `xtreg, re`.

Interpreting the Breusch–Pagan Result

Fail to reject H_0

No evidence of a unit effect. Pooled OLS is a defensible, simpler choice — no need for RE or FE.

Reject H_0

A real unit effect is present. Pooling is misspecified — proceed to the Hausman test to choose RE vs. FE.

In most finance and management panels this test rejects H_0: heterogeneity across firms is the norm, not the exception.

The Hausman Test, Formally

Compare $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$, scaled by the variance of their difference:

$$H = (\hat{\beta}^{FE} - \hat{\beta}^{RE})' \left[\widehat{\text{Var}}(\hat{\beta}^{FE}) - \widehat{\text{Var}}(\hat{\beta}^{RE}) \right]^{-1} (\hat{\beta}^{FE} - \hat{\beta}^{RE}) \sim \chi^2_k$$

H_0: $\text{Cov}(\alpha_i, x_{it}) = 0$ (RE consistent and efficient) $\quad$ H_1: RE inconsistent, FE consistent

1. **k degrees of freedom** — One per *time-varying* slope coefficient compared across the two models
2. **Built on the efficiency gap** — Under H_0 RE is efficient, so $\text{Var}(\hat{\beta}^{FE} - \hat{\beta}^{RE}) = \text{Var}(\hat{\beta}^{FE}) - \text{Var}(\hat{\beta}^{RE})$
3. **Large H = the estimates disagree** — Disagreement is the sign that RE's assumption is failing

What the Hausman Test Requires

1	**RE must be efficient under H_0**	That is what justifies the simple variance difference in the formula
2	**FE must be consistent under both hypotheses**	FE gives the right answer whether or not RE's assumption holds (it needs strict exogeneity)
3	**Classical (non-robust) standard errors**	RE is only efficient with homoskedastic, serially uncorrelated ε_{it} — rarely true in practice
4	**A well-behaved variance difference**	$\text{Var}(\hat{\beta}^{FE}) - \text{Var}(\hat{\beta}^{RE})$ must be positive semi-definite — not guaranteed in finite samples

When the Hausman Test Misbehaves

A negative test statistic

Possible in finite samples when the variance difference isn't positive definite. Treat it as a warning sign, not as "fail to reject".

Clustered or robust SEs

The classical test is invalid — Stata's `hausman` refuses to run after `vce(cluster)` or `vce(robust)`.

Little within-unit variation

If FE is imprecise to begin with, the test has little power: "fail to reject" may just mean "can't tell".

The fix: a robust version

Use `xtoverid` after a clustered RE model, or the Mundlak approach (next slide).

A Robust Alternative: Mundlak's Correlated Random Effects

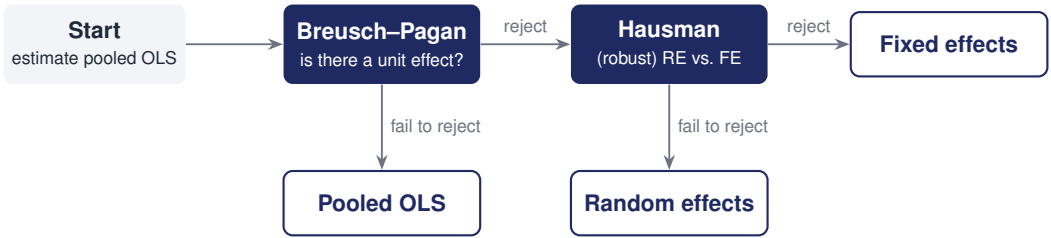
Add each unit's time averages of the regressors to the RE model (Mundlak, 1978):

$$y_{it} = \beta_0 + \beta_1 x_{it} + \gamma\, \bar{x}_i + \alpha_i + \varepsilon_{it}$$

1. **$\bar{x}_i$ soaks up the correlation** — Any part of α_i correlated with x is modeled through $\gamma \bar{x}_i$
2. **$\hat{\beta}_1$ equals the FE estimate** — You get the within effect — and can still include time-invariant regressors
3. **Testing $\gamma = 0$ is a Hausman test** — A simple Wald test that works with clustered standard errors

$\gamma = 0$ means RE's assumption holds; rejecting it favors FE. See Wooldridge (2010, ch. 10).

The Complete Testing Sequence



In practice, also report the *F*-test that all α_i are equal (bottom of `xtreg, fe` output). Tests inform the choice, but your **identification argument** should drive it.

Breusch–Pagan and Hausman in Stata

```
clear all
webuse nlswork
xtset idcode year
xtreg ln_wage hours union age, re   // classical SEs: needed for hausman
estimates store re_model
xttest0                             // Breusch-Pagan LM: pooled OLS vs. RE
xtreg ln_wage hours union age, fe
estimates store fe_model
hausman fe_model re_model           // consistent model first, efficient second
```

`xttest0`	Breusch–Pagan LM test; must follow `xtreg, re`
`estimates store`	saves each model's results under a name
`hausman`	compares the stored models; order matters (consistent, then efficient)

Robust Hausman Tests in Stata

Option 1: `xtoverid`

```
ssc install ivreg2
ssc install ranktest
ssc install xtoverid
xtreg ln_wage hours union age, ///
    re vce(cluster idcode)
xtoverid
```

User-written; reports a robust test of RE vs. FE.

Option 2: Mundlak by hand

```
quietly xtreg ln_wage hours union age, re
generate byte insample = e(sample)
foreach v of varlist hours union age {
  bysort idcode: egen m_‘v’ = mean(‘v’) if insample
}
xtreg ln_wage hours union age m_*, ///
    re vce(cluster idcode)
test m_hours m_union m_age
```

Means computed on the estimation sample; `test` is the robust Hausman.

So Far: Choosing the Right Estimator

Breusch–Pagan and Hausman together give you a defensible, testable path from pooled OLS to random or fixed effects.

But even a well-chosen unit-effects model can miss something: **shocks that hit every unit at the same time**.

A financial crisis, a new regulation, a pandemic, a change in the Selic rate — these move every firm in the same period, regardless of firm-specific traits.

That's what time fixed effects are for.

What Are Time Fixed Effects?

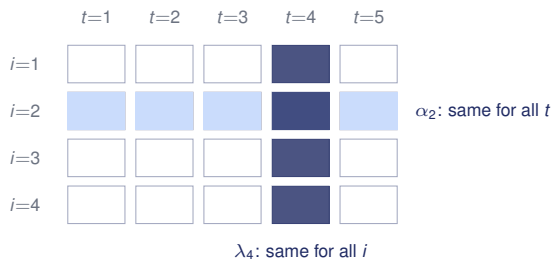
Time fixed effects control for anything that varies across periods but is **common to every unit** in that period — the mirror image of unit fixed effects.

Unit FE (α_i)

Absorb whatever is stable about a firm across all the years it is observed — one per *row*

Time FE (λ_t)

Absorb whatever is common to all firms in a given year — one per *column*



The Two-Way Fixed Effects Model

$$y_{it} = \beta_1 x_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

α_i absorbs unit heterogeneity; λ_t absorbs period-specific shocks common to all units. Only what is left — variation specific to unit *and* period — identifies β_1.

In a **balanced** panel, the two-way within transformation is

$$\ddot{y}_{it} = y_{it} - \bar{y}_i - \bar{y}_t + \bar{y}$$

The grand mean $\bar{y}$ is added back because it was subtracted twice. In **unbalanced** panels this shortcut fails: use year dummies or iterative demeaning (`reghdfe`).

Why Add Time Effects: Real Examples

A financial crisis

2008–2009 depressed ROA for almost every firm, regardless of governance quality

A regulatory change

A new disclosure rule shifts every listed firm's reporting behavior in the same year

Macroeconomic cycles

Interest-rate cycles (e.g., the Selic) move firms' financing decisions together, market-wide

Common technology shocks

A widely adopted innovation (e.g., cloud computing, Pix) affects most firms in the same period

If x trends over time and y does too, omitting λ_t attributes the common trend to x.

Estimating Time Effects: The Mechanics

1	**Add one dummy per period**	Like unit dummies in LSDV, but indexed by t: $T - 1$ dummies, to avoid perfect collinearity with the intercept
2	**Or demean on both dimensions**	Subtract unit and period means and add back the grand mean (balanced panels only)
3	**Equivalent results either way**	Both recover the identical $\hat{\beta}_1$, just as with unit-only fixed effects
4	**Absorbs common trends and level shifts**	Any period-specific shift in mean y is soaked up by λ_t, not attributed to x
5	**Absorbs common regressors too**	Anything that varies only over time (Selic, GDP growth) can no longer be estimated

Reading a Two-Way Fixed Effects Coefficient

"Holding this unit's own traits and this period's common shocks fixed, how does y move with x?"

1	**Purged of unit heterogeneity**	Whatever is stable about the unit no longer contaminates $\hat{\beta}_1$
2	**Purged of common shocks**	Whatever hit every unit in the same period no longer contaminates $\hat{\beta}_1$ either
3	**Cleaner — but still not causal**	Confounders that vary by both unit and time can still bias the result

Looking ahead: when *x* is a treatment adopted at different times, TWFE has known problems (Goodman-Bacon, 2021). We return to this with difference-in-differences.

Testing Whether Time Effects Are Needed

Test whether the period dummies matter *jointly*, rather than relying on individual t-statistics.

$$H_0\colon \lambda_2 = \lambda_3 = \cdots = \lambda_T = 0 \quad \text{(no time effects needed)}$$

A joint Wald (F) test on all the time dummies, run after estimating the two-way model.

Reject $H_0 \Rightarrow$ keep the time effects; otherwise common shocks may bias the coefficient on x.
In practice, year effects are cheap insurance in short panels, and most referees will expect them.

Two-Way Fixed Effects in Stata

```
clear all
webuse nlswork
xtset idcode year
xtreg ln_wage hours union age i.year, fe vce(cluster idcode)
testparm i.year    // do the year dummies matter jointly?
```

`i.year`	factor-variable notation: one dummy per year (the time fixed effects)
`testparm i.year`	joint Wald test that all year coefficients are zero

In class: is anything reported as *omitted*? Why might `age` be a problem once you include both individual and year fixed effects?

What If I Don’t Want Hundreds of Dummies in My Output?

```
ssc install ftools
ssc install reghdfe
reghdfe ln_wage hours union age, absorb(idcode year) vce(cluster idcode)
```

`reghdfe`	estimates the FE model by iterative demeaning, without estimating dummy coefficients (Correia, 2016)
`absorb()`	lists the fixed effects to absorb — any number of dimensions (firm, year, industry$\times$year, . . .)

`reghdfe` drops **singletons** (units observed only once) by default, so N and SEs can differ slightly from `xtreg, fe`. Same $\hat{\beta}$ otherwise.

Check Your Understanding

1. **Mixed signals**
   Breusch–Pagan rejects; the classical Hausman test gives $p = 0.40$. Which estimator do you choose, and what caveat do you add?

2. **Stata error**
   Both models were estimated with `vce(cluster)` and `hausman` refuses to run. What do you do instead?

3. **Time effects**
   Your sample of Brazilian listed firms covers 2006–2012. Give one concrete reason to include year FE.

4. **What's left?**
   Which regressors can no longer be estimated in a two-way FE model? Give one example of each kind.

Recap and What's Next

Takeaway: Breusch–Pagan and (robust) Hausman tests give you a tested path to the right estimator; time fixed effects add a second layer of protection against confounders shared across units in the same period.

Next: Hour 3 — Paper Discussions

Three papers on different subjects, each using different panel data techniques.

Questions?

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